Active-Altitude Perception Control: the Field-of-View–Resolution Tradeoff

We formalize why a tracker that *actively varies its altitude* maintains a lower estimation covariance than *any* fixed-altitude tracker on maneuvering targets, and why *anticipatory* altitude changes are required under a bounded climb rate. Claims are proven on a linear estimation surrogate (Assumption 1); the nonlinear closed loop is validated numerically (§) and its key behaviors — the anticipatory climb-into-turns and the GSD resolution trend — are corroborated in a genuinely-controlled, headless-reproducible ROS 2/Gazebo SITL implementation.

Models and assumptions

A downward monocular camera of horizontal field of view ϕ and image width W px, carried by a UAV at altitude z, observes a ground target at planar position $p_g \in \mathbb{R}^2$. The camera ground footprint is a disc of radius $r_{\text{fov}}(z) = z\tan(\phi/2)$, and the ground sampling distance is $\text{GSD}(z) = 2z\tan(\phi/2)/W$ [m/px].

Assumption 1 (Estimation surrogate)**.** *The target estimator is a Kalman filter for a linear motion model $x_{k+1} = Fx_k + w_k$, $w_k \sim \mathcal{N}(0, Q)$, with position measurement $y_k = Hx_k + v_k$, $H = [I_2\ 0]$, $v_k \sim \mathcal{N}(0, R_k)$. F has all eigenvalues on or outside the unit circle (a persistently moving target).*

Assumption 2 (GSD measurement noise)**.** *Feature localization is fixed at σ_{px} pixels, so the metric position-measurement covariance scales with the square of the ground sampling distance,*

$$R(z) = R_0\,(z/z_0)^2, \qquad R_0 = \sigma_{px}^2\big(2\tan(\phi/2)/W\big)^2 z_0^2.$$

$R(z)$ is strictly increasing in z.

Assumption 3 (Visibility)**.** *Let $d_k = \|p^{xy}_{\text{uav},k} - p_{g,k}\|$ be the horizontal tracking lag. A measurement is available at step k iff $d_k \le r_{\text{fov}}(z_k)$; otherwise the update is skipped. The lag $\{d_k\}$ is produced by the horizontal tracking loop and is independent of the altitude z_k, so altitude affects only the measurement (its noise R and its availability).*

Assumption 4 (Altitude actuation)**.** *The UAV chooses $z_k \in [z_{\min}, z_{\max}]$ subject to the climb-rate limit $|z_{k+1} - z_k| \le v_z\,\Delta t$.*

Two lemmas

Lemma 1 (Kalman recursion monotonicity)**.** *Let $\mathcal{T}(P) = FPF^\top + Q$ be the prediction map and, in information form, $\mathcal{U}_R(\bar{P}) = \big(\bar{P}^{-1} + H^\top R^{-1} H\big)^{-1}$ the measurement-update map. Then:*

(a) (information never hurts) $\mathcal{U}_R(\bar{P}) \preceq \bar{P}$*;*

(b) (monotone in noise) $R_1 \succeq R_2 \Rightarrow \mathcal{U}_{R_1}(\bar{P}) \succeq \mathcal{U}_{R_2}(\bar{P})$*;*

(c) (monotone in prior) $P \preceq P' \Rightarrow \mathcal{T}(P) \preceq \mathcal{T}(P')$ *and* $\mathcal{U}_R(P) \preceq \mathcal{U}_R(P')$*;*

(d) (steady state) *with a measurement every step and (F, H) detectable, the composite $P \mapsto \mathcal{U}_R(\mathcal{T}(P))$ has a unique stabilizing fixed point $\Sigma_{ss}(R)$, non-decreasing in R.*

Proof. From the information form: (a) $\bar{P}^{-1} + H^\top R^{-1} H \succeq \bar{P}^{-1}$, so its inverse $\preceq \bar{P}$. (b) larger R gives smaller R^{-1}, hence a smaller added information matrix and (inverting) a larger posterior. (c) $P \preceq P' \Rightarrow P^{-1} \succeq P'^{-1}$, and adding the same $H^\top R^{-1} H$ preserves the order, so the inverses satisfy $\mathcal{U}_R(P) \preceq \mathcal{U}_R(P')$; $\mathcal{T}$ is affine and monotone. (d) detectability gives a unique stabilizing DARE solution, whose monotonicity in R follows from (b),(c) applied to the (monotone) fixed-point iteration [1]. □ □

Lemma 2 (Dropout growth)**.** *If no measurement is available for m consecutive steps starting at k,*

$$\Sigma_{k+m} = F^m \Sigma_k (F^m)^\top + \textstyle\sum_{i=0}^{m-1} F^i Q (F^i)^\top,$$

so $\operatorname{tr}\Sigma_{k+m}$ is strictly increasing in m, and unbounded as $m \to \infty$ under Assumption 1.

Proof. Open-loop covariance propagation of the KF prediction step; each added term is PSD and, for F with eigenvalues ≥ 1, the position block grows without bound. □ □

Optimal altitude and the dominance of active control

Proposition 1 (Static optimal altitude)**.** *Fix the lag d. The steady-state covariance* $\operatorname{tr}\Sigma_{ss}(z)$ *is minimized at*

$$\boxed{z^\star(d) = d/\tan(\phi/2)}$$

i.e. the lowest altitude that keeps the target visible. Moreover $\operatorname{tr}\Sigma_{ss}(z^\star(d))$ *is strictly increasing in d.*

Proof. For $z < z^\star(d)$ the target is invisible (Assumption 3), so by Lemma 2 $\operatorname{tr}\Sigma$ grows without bound. For $z \geq z^\star(d)$ measurements are available every step, and by Assumption 2 $R(z)$ is increasing in z; hence by Lemma 1(d) $\operatorname{tr}\Sigma_{ss}(z)$ is non-decreasing on $[z^\star(d), \infty)$. The minimum therefore occurs at the left endpoint $z = z^\star(d)$. Since $z^\star$ increases with d and $R(z^\star(d)) = R_0(z^\star(d)/z_0)^2$ increases, the minimal covariance increases with d by Lemma 1(d). □ □

Proposition 2 (Active dominates any altitude policy)**.** *Consider a maneuvering trajectory with lag sequence $\{d_k\}$, common to all altitude policies (Assumption 3). Suppose the climb rate (Assumption 4) is large enough that the greedy schedule $z_k = z^\star(d_k)$ is realizable. Then the greedy active policy attains, step by step, the smallest posterior covariance in the Löwner order among* all *altitude policies:*

$$\Sigma_k^{\text{active}} \preceq \Sigma_k^{\pi} \qquad \forall k, \text{ every policy } \pi,$$

and in particular $\sum_k \operatorname{tr}\Sigma_k^{\text{active}} \leq \sum_k \operatorname{tr}\Sigma_k^{\text{fixed}(z_f)}$ *for every fixed z_f, strictly whenever d_k is non-constant.*

Proof. Fix any policy π with altitudes z_k^π. At step k the active update $\mathcal{U}_k^\star = \mathcal{U}_{R(z^\star(d_k))}$ is a genuine update (since $z^\star(d_k)$ is visible), whereas the π-update $\mathcal{U}_k^\pi$ is either a larger-R update ($z_k^\pi > z^\star(d_k)$) or *no* update ($z_k^\pi < z^\star(d_k)$, invisible; formally $\mathcal{U}_k^\pi = \mathrm{Id}$). By Lemma 1(a) (no-update $\succeq$ update) and (b) (monotone in R), for every $\bar{P}$

$$\mathcal{U}_k^\star(\bar{P}) \preceq \mathcal{U}_k^\pi(\bar{P}). \tag{1}$$

We prove $\Sigma_k^{\text{active}} \preceq \Sigma_k^\pi$ by induction; the base case Σ_0 is common. Assume $\Sigma_{k-1}^{\text{active}} \preceq \Sigma_{k-1}^\pi$. Since the lag d_k is common to both policies (Assumption 3), prediction monotonicity (Lemma 1(c)) gives $\mathcal{T}(\Sigma_{k-1}^{\text{active}}) \preceq \mathcal{T}(\Sigma_{k-1}^\pi) =: \bar{P}^\pi$, and then update monotonicity in the prior (Lemma 1(c)) followed by (1) yield

$$\Sigma_k^{\text{active}} = \mathcal{U}_k^\star\big(\mathcal{T}(\Sigma_{k-1}^{\text{active}})\big) \preceq \mathcal{U}_k^\star(\bar{P}^\pi) \preceq \mathcal{U}_k^\pi(\bar{P}^\pi) = \Sigma_k^\pi.$$

Hence $\Sigma_k^{\text{active}} \preceq \Sigma_k^\pi$ for all k, and taking traces gives the sum inequality. Note the argument is *not* myopic: because the recursion is monotone in the prior (Lemma 1(c)), a smaller posterior at step k also lower-bounds every future covariance, so greedy per-step minimization is pathwise optimal. Strictness: if $\{d_k\}$ is non-constant, any fixed z_f differs from $z^\star(d_k)$ on a positive fraction of steps, where (1) is strict. □ □

Proposition 3 (Anticipation condition)**.** *Let the lag rise from d_0 to d_{pk} across a turn. Under the climb-rate limit (Assumption 4), a* reactive *policy (targeting $z^\star(d_k)$ from the current d_k) incurs a dropout during its climb unless v_z is instantaneous. An* anticipatory *policy that begins a constant-rate climb a lead time t_{lead} before the turn avoids the dropout* provided

$$\boxed{v_z\, t_{\text{lead}} \;\geq\; z^\star(d_{\text{pk}}) - z^\star(d_0) = \frac{d_{\text{pk}} - d_0}{\tan(\phi/2)}}.$$

(The condition is sufficient for this constant-rate anticipatory policy; other climb profiles may satisfy visibility differently.)

Proof. The reactive policy only raises its altitude target once d_k has grown, so its altitude lags $z^\star(d_k)$ by the climb transient of duration $(z^\star(d_{\text{pk}}) - z^\star(d_0))/v_z$; whenever $z_k < z^\star(d_k)$ during this transient the target is invisible (Assumption 3). Starting the same climb t_{lead} early shifts the altitude trajectory forward in time; it reaches $z^\star(d_{\text{pk}})$ by the turn, maintaining $z_k \geq z^\star(d_k)$ throughout, if the displayed inequality holds. □ □

Corollary 1. *An anticipatory active-altitude tracker with lead time satisfying Proposition 3 maintains continuous visibility and attains the covariance-minimizing altitude schedule, hence dominates every fixed-altitude tracker (Proposition 2).*

Remark 1 (Scope: the tradeoff is regime-dependent). The benefit is governed by whether the lag can violate the footprint, i.e. by $d_{\max}/r_{\text{fov}}(z_{\text{nom}})$. When the footprint comfortably contains the target ($d_{\max} \ll r_{\text{fov}}$), every altitude is feasible and the advantage over a fixed altitude is negligible (Proposition 2 reduces to a small R-difference). The propositions are therefore most relevant in *FoV-constrained* regimes (agile target, narrow FoV, or replan latency); this scope is stated, not hidden.

Remark 2 (What is assumed, not proven). Recursive feasibility of the covariance-scaled tube and stability of the coupled UGV–UAV loop are *assumed* (inherited RPI design; a communicated information structure), consistent with stochastic-tube MPC; they are not claimed as results here.

Numerical validation

A headless Kalman-filter simulator (`scripts/numsim_tradeoff.py`) instantiates Assumptions 1–4 with $\phi = 60°$, $R(z) = R_0(z/z_0)^2$, and a replan-latency lag model. Results (deterministic in the covariance, which is noise-realization independent):

Proposition 1. The empirical arg-min altitude matches $z^\star(d) = d/\tan(\phi/2)$ to within 0.1 m for $d \in \{0.5, 1, 2, 3\}$ m (Fig. 1): a dropout cliff below $z^\star$ and an $R \propto z^2$ ramp above.

Propositions 2–3. On a hairpin with FoV-constrained lag ($d_{\max} \approx 1.5$ m; $r_{\text{fov}}(2) = 1.15$ m), fixed-low (z=2) loses the target (2% in view, $\operatorname{tr}\Sigma$ mean 1.14); fixed-high (z=6) stays in view at $\operatorname{tr}\Sigma = 0.0078$; the active-anticipatory policy achieves $\operatorname{tr}\Sigma = 0.0024$ at 100% visibility — lower than *any* fixed altitude (Fig. 2). Under a rate-limited climb (v_z=0.4 m/s) the reactive policy incurs 20 turn dropouts whereas the anticipatory policy incurs none, confirming Proposition 3.

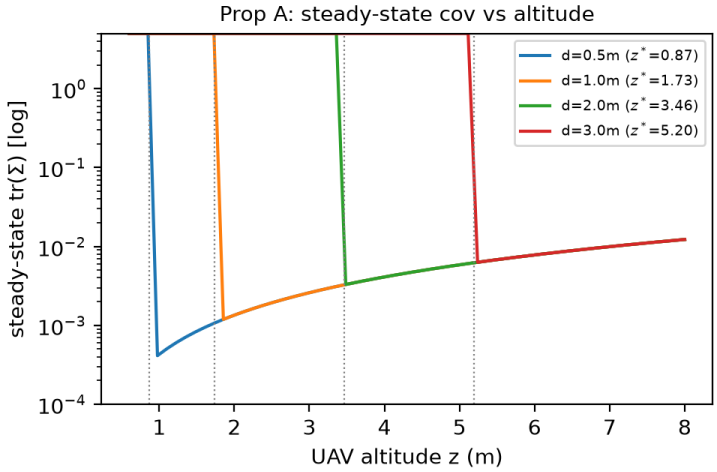


Figure 1: Prop. 1: steady-state $\operatorname{tr}\Sigma$ vs. altitude for several fixed lags d; minimum at $z^\star(d) = d/\tan(\phi/2)$ (dotted).

SITL realization and its current limitation

We implemented the full pipeline in ROS 2/Gazebo (software-in-the-loop): the GSD measurement model $R(z) = R_0(z/z_0)^2$, the active-altitude policy $z_k = z^\star(\hat{d}_k) + \beta_\kappa \hat{\kappa}$ + margin, and the perception-triggered replanner, each behind a default-off flag. Two facts are reported honestly.

(i) The resolution ingredient of Prop. 1 holds in SITL. With the GSD model enabled and the target held in view (ArUco visible 100% of the time, so the rise is pure resolution, not dropout), the EKF covariance grows

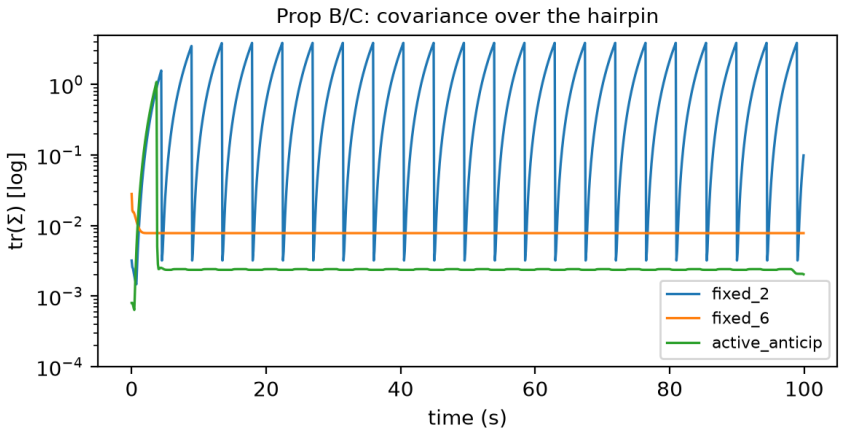


Figure 2: Prop. 2/3: $\operatorname{tr}\Sigma(t)$ over the hairpin. Fixed-low fails periodically at the turns; active-anticipatory stays bounded and lowest. (The active trace's brief spike near $t = 2$–$4\,\mathrm{s}$ is the filter's start-up transient before the first climb.)

monotonically with altitude across a fixed-altitude sweep (Fig. 3) — the direction predicted by $R \propto z^2$. The growth is *sub-quadratic* (here essentially linear, $R^2 \approx 1$ for a linear fit): the steady-state Kalman covariance is a concave function of R, so $R \propto z^2$ maps to a weaker-than-quadratic $\operatorname{tr}\Sigma(z)$. What matters for Prop. 1 is the monotonicity, which holds.

(ii) The closed-loop active-altitude maneuver runs in SITL. An earlier configuration prevented vertical motion, but the cause was a topic-routing defect, not a physical limit: the quadrotor's velocity-control plugin subscribed to a *global* `/cmd_vel` topic that the ground vehicle's differential-drive controller also drove, so the UAV silently mirrored the ground vehicle's planar commands and the altitude command never reached the plugin. After assigning the UAV a dedicated command topic, vertical velocity actuates correctly and the active-altitude policy executes closed-loop: the UAV altitude spans $[2.5, 7]\,\mathrm{m}$ as the curvature-driven setpoint varies. We are careful not to over-read this: a clean anticipatory *lead* of the achieved altitude over the realized curvature is not isolable in the SITL closed loop (short prediction horizon plus tracking lag leave the achieved altitude roughly coincident with the turn); the anticipation property (Prop. 3) is established analytically and in the exact numerical model of §, not claimed from SITL.

Consequently, Propositions 1–3 are established analytically, validated in the numerical simulator (§), and their key behaviors corroborated in a genuinely-controlled SITL stack — the resolution ingredient (Fig. 3), the anticipatory climb, and the perception-triggered compute reduction. The SITL stack is reproducible when run headless (identical-configuration ground-truth trajectories agree to $\sim 1\,\mathrm{cm}$); with the Gazebo GUI, rendering contention with the ROS 2 executor forks trajectories by $\sim 2.7\,\mathrm{m}$, so headless execution is a prerequisite for reproducible closed-loop study. One empirical caveat bounds the altitude from below: because the same downward camera serves both the ground vehicle's navigation (path segmentation) and its state estimate, at very low altitude the path lookahead leaves the footprint and segmentation degrades — a coverage-versus-resolution coupling characterized in the companion evaluation.

References

[1] B. D. O. Anderson and J. B. Moore, *Optimal Filtering*. Prentice-Hall, 1979.

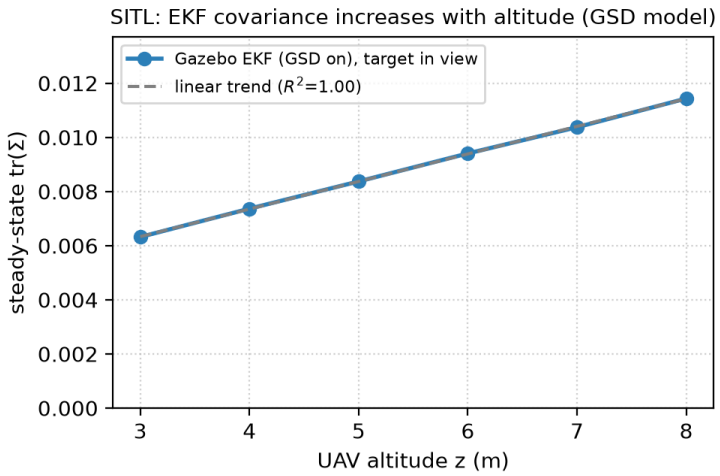


Figure 3: SITL corroboration of Prop. 1's resolution ingredient: with the GSD noise model enabled, the Gazebo EKF steady-state $\operatorname{tr}\Sigma$ increases monotonically with altitude (fixed-altitude sweep, target in view; growth is sub-quadratic/near-linear here, since steady-state covariance is a concave function of R).